Title: SOC Alert Management and Incident Response – Week 2

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Task: SOC Alert Management and Incident Response

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1. Introduction

This document presents the implementation of Security Operations Center (SOC) activities including alert management, incident classification, alert triage, evidence preservation, and incident response. The objective of this task is to simulate real-world SOC operations using structured methodologies and tools. The document demonstrates how alerts are analyzed, prioritized, validated, and handled efficiently to mitigate potential security threats.

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2. Alert Management

2.1 Alert Classification Table

The following table represents different types of security alerts categorized based on priority and mapped with MITRE ATT&CK techniques.

Alert ID	***Type***	***Priority***	***MITRE***
1	*Phishing*	*High*	*T1566*
2	*Ransomware*	*Critical*	*T1486*
3	*Port Scan*	*Medium*	*T1046*
4	*SSH Brute Force*	*High*	*T1110*
5	*Malware Infection*	*High*	*T1204*

6	*Privilege Escalation*	*Critical*	*T1068*
7	*DDoS Attack*	*High*	*T1498*
8	*Suspicious Login*	*Medium*	*T1078*
9	*Data Exfiltration*	*Critical*	*T1041*
10	*Command Execution*	*High*	*T1059*

(Caption: Figure 1 – Alert Classification and Prioritization Table)

2.2 Incident Ticket

Title: Critical Alert – Ransomware Detected on Server-X

Date: 20 March 2026

Time: 14:00 IST

Description:

A ransomware attack has been detected on Server-X. A suspicious file named *crypto_locker.exe* was identified. Unauthorized encryption activity was observed, indicating potential system compromise.

Indicators of Compromise (IOCs):

- File: crypto_locker.exe
- IP Address: 185.199.108.153
- File Hash: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

Priority: Critical

Status: Escalated

Assigned To: SOC Analyst

Action Taken:

- System isolated from network
- Malicious process identified and contained
- Initial forensic analysis initiated

2.3 Escalation Email

Subject: Critical Alert Escalation – Ransomware Detected

Dear Tier 2 Team,

A critical ransomware incident has been identified on Server-X. The system shows active encryption behavior linked to crypto_locker.exe. The source IP 185.199.108.153 has been flagged as suspicious. Immediate containment actions have been initiated, including isolating the affected system.

Further investigation is required to assess the impact and prevent lateral movement. Kindly prioritize this incident and proceed with advanced analysis.

Regards,

SOC Analyst

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3. Alert Triage and IOC Validation

3.1 Alert Triage Table

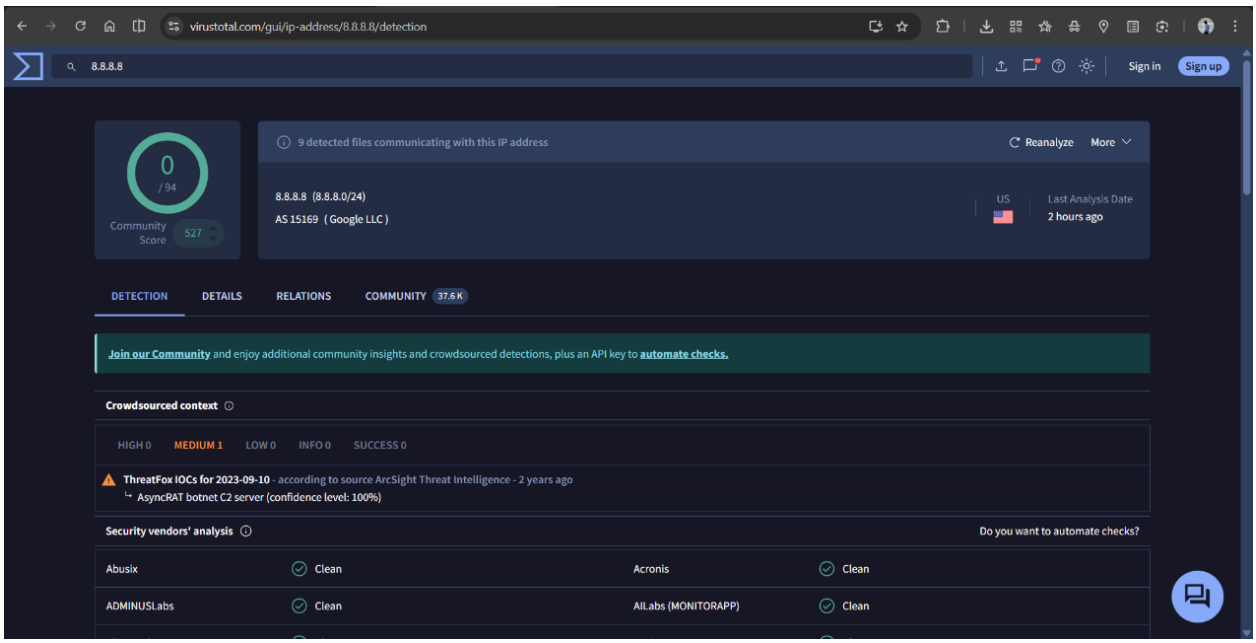
The following table shows alert analysis with priority and investigation status.

Alert ID	Description	IP	Priority	Status
2	SSH Brute Force	192.168.1.100	Medium	Open
3	Port Scan Detected	192.168.1.105	Low	Open
4	Suspicious Login	10.0.0.5	High	Investigating
5	Malware Traffic	8.8.8.8	High	Open
6	Failed Login Attempts	185.199.108.153	Medium	Open

(Caption: Figure 2 – Alert Triage and Status Tracking)

3.2 IOC Validation

The IP address was analyzed using VirusTotal to determine its reputation and potential malicious activity.



(Caption: Figure 3 – IOC Validation using VirusTotal)

Summary:

The IP address was analyzed using VirusTotal. The results indicate that most security vendors marked the IP as clean, with minimal suspicious activity reported. This suggests a low risk; however, continuous monitoring is recommended to ensure network security and prevent potential threats.

4. Evidence Preservation

The following table documents the collected digital evidence and associated hash values to ensure data integrity.

Item	Description	Date	Hash
Evidence File	evidence.txt	20 March 2026	fb1e17c2023b5d596bbecebbd7ebc6186e052a2bc230a7683f4b66f060cb4839
Memory Dump	Server-X RAM capture	20 March 2026	SHA256: abc123xyz456
Disk Image	Server-X Disk Backup	20 March 2026	SHA256: def456uvw789
Log File	Security Logs	20 March 2026	SHA256: ghi789rst012
Network Capture	Packet Capture (PCAP)	20 March 2026	SHA256: jkl012mno345

(Caption: Figure 4 – Evidence Collection and Hash Values)

5. Capstone Project – Incident Response

5.1 Detection Log

Time	IP	Alert	MITRE
11:00	192.168.1.100	Web Exploit	T1190
11:05	192.168.1.100	Brute Force	T1110
11:10	192.168.1.100	Suspicious Activity	T1046

5.2 Response Actions

- Isolated affected system
- Blocked malicious IP address
- Initiated investigation

5.3 Incident Report

A simulated cyber attack was conducted to test the detection and response capabilities of the SOC environment. The attack involved an exploit attempt originating from IP address 185.199.108.153 targeting Server-X. The alert was successfully detected and classified as critical due to its potential impact on system integrity.

Upon detection, immediate response actions were taken, including isolating the affected system from the network and blocking the malicious IP address. Indicators of compromise were analyzed, and evidence was collected to support further investigation. The response team ensured that the threat was contained effectively without causing major damage.

The incident highlights the importance of real-time monitoring, rapid response, and proper documentation in SOC operations. Continuous improvement in detection mechanisms and staff training is recommended to enhance security posture and reduce response time in future incidents.

5.4 Manager Brief

A security incident involving a potential cyber attack was detected and handled successfully. The affected system was quickly isolated, and the source of the attack was blocked to prevent further impact. No major damage was observed, and the situation was brought under control efficiently.

This incident demonstrates the effectiveness of current monitoring and response processes. However, improvements in early detection and preventive measures are recommended to strengthen overall security. Continuous monitoring and staff awareness will help reduce future risks and ensure system safety.

6. Conclusion

This project demonstrates the implementation of SOC operations including alert management, triage, incident response, and evidence handling. The simulation highlights the importance of structured processes and tools in identifying and mitigating threats. Proper documentation and response strategies are essential for maintaining cybersecurity resilience.

7. References

- MITRE ATT&CK Framework
- VirusTotal
- NIST SP 800-61 Incident Handling Guide
- SANS Incident Handler’s Handbook